

Merging the log families of the reduced cutoff expansion

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Abstract

The reduced $d = 2$ cutoff expansion listed three logarithmic families: the u -face logs, the v -face logs, and the subtracted corner logs. All three are supported on the collision predicate $(h_1 + i + 1)/k_1 = (h_2 + j + 1)/k_2$, and for fixed i (resp. j) at most one partner collides. Hence the v -face family equals the corner family exactly and the reduced sum equals a merged sum with ONE transferred log term $N^{-\alpha_i}(M[\alpha_i; 0; c_{ij}/(k_1 k_2)] \log N - M[\alpha_i; 1; c_{ij}/(k_1 k_2)])$ per collision (i, j) (`reducedSum_eq_merged`). The proof is integrability-free. The analytic cutoff theorems are restated in merged form. Zero sorry/axiom.

1 The things formalised

```
theorem transferTerm_ite ( : ) (n : ) (P : Prop) [Decidable P] (g : → ) (N : ) :
  transferTerm n (fun s => if P then g s else 0) N
  = if P then transferTerm n g N else 0
theorem transferTerm_sum_ite_unique { : Type*} ( : ) (n : ) (t : Finset )
  (Q : → Prop) [DecidablePred Q] (huniq : i t, j t, Q i → Q j → i = j)
  (g : → → ) (N : ) :
  transferTerm n (fun x => i t, if Q i then g i x else 0) N
  = i t, if Q i then transferTerm n (g i) N else 0
theorem uface_resonance_iff (h h k k i m : ) (hk : 0 < k) :
  ((m : ) = (k : ) * (((h + i : ) : ) + 1) / k) - h - 1
  (((h + i : ) : ) + 1) / k = (((h + m : ) : ) + 1) / k
noncomputable def mergedLog ( : ) (h h k k M M : ) (c : → → → )
  (N : ) : :=
  i Finset.range M, j Finset.range M,
  if (((h + i : ) : ) + 1) / k = (((h + j : ) : ) + 1) / k then
    transferTerm (((h + i : ) : ) + 1) / k 1
    (fun s => c i j s / ((k : ) * k)) N
  else 0
noncomputable def mergedSum ... :=
  (u-face finite parts) + (v-face finite parts) + mergedLog
theorem reducedSum_eq_merged ( b : ) (h h k k M M : ) (hk : 0 < k)
  (hk : 0 < k) (a bj : → → → ) (c : → → → ) (N : ) :
  reducedSum b h h k k M M a bj c N = mergedSum b h h k k M M a bj c N
theorem twoD_cutoff_analytic_merged ... -- twoD_cutoff_analytic with mergedSum
theorem twoD_cutoff_amplitude_merged ... -- twoD_cutoff_amplitude with mergedSum
```

2 Mathematics

The three resonance predicates $m = k_2 \alpha_i - h_2 - 1$ (u -face), $m = k_1 \delta_j - h_1 - 1$ (v -face) and $0 = k_1 \delta_j - (h_1 + i) - 1$ (corner) are all equivalent to the collision $\alpha_i = \delta_j$ with $\alpha_i = (h_1 + i + 1)/k_1$,

$\delta_j = (h_2 + j + 1)/k_2$. The u -face log coefficient at i is the function $s \mapsto \sum_{j < M_2} [\alpha_i = \delta_j] c_{ij}(s)/(k_1 k_2)$; since $j \mapsto \delta_j$ is injective, at most one j contributes, so the transferred term of this sum of indicators equals the sum of indicator-weighted transferred terms (a two-case argument: a unique witness, or none, in which case both sides vanish because `transferTerm` of the zero function is zero). The same applies to the v -face family, which after `Finset.sum_comm` coincides term by term with the corner family (whose coefficient `faceLogCoeff` of `cornerData` is a single indicator). So $U + V - C = U$, and U is the merged family. On a collision $\alpha_i = \delta_j$, so the exponent may be written either way. Astra #8's monomial tests hold for the merged form: with $M[\alpha; \ell; f] = \int_0^\infty s^{\alpha-1} (\log s)^\ell e^{-\beta s^2} f(s) ds$ the collision term is $N^{-\alpha} (M[\alpha; 0; f] \log N - M[\alpha; 1; f]) / (k_1 k_2)$ (sign fixed by $\log(s/N) = \log s - \log N$), the two face finite parts add $(k_1 + k_2) \log b M[\alpha; 0; f] / (k_1 k_2)$ at a collision with $b \neq 1$, and off collisions the faces give $b^{k_2(d-a)} N^{-a} M[a; 0; f] / (k_1 k_2 (d-a)) - b^{k_1(a-d)} N^{-d} M[d; 0; f] / (k_1 k_2 (d-a))$.

3 Paper-facing meaning

This is the “decisive bookkeeping lemma” of Astra #8: the reduced sum is now in the normal form $\sum_i N^{-\alpha_i} M[\alpha_i; 0; U] + \sum_j N^{-\delta_j} M[\delta_j; 0; V] + \sum_{\text{collisions}} N^{-\alpha_i} (M[\alpha_i; 0; C] \log N - M[\alpha_i; 1; C])$ from which the $d = 2$ Taylor-tree statement (one polynomial P_μ of degree ≤ 1 in $\log N$ per pole μ , coefficients independent of the cutoff) is a regrouping by exponent. Next: cutoff independence of the finite parts and canonical $U_\alpha, V_\alpha, C_\alpha$, then the wrapper `twoD_taylor_tree_equal`.

4 Lean notes

- `transferTerm_zero` already existed with a different meaning (log-degree 0); the zero-function lemma is `transferTerm_zero_fun`, following `logMoment_zero_fun`.
- `if_congr (hiff) (by ring) rfl` rewrites a decidable condition inside `ite` together with its branch; `Finset.sum_congr rfl fun j _ => if_congr ...` handles the indicator sums without `simp` normalising the real equations.
- Injectivity of $i \mapsto (h_1 + i + 1)/k_1$: `div_left_inj'`, `push_cast`, then `exact_mod_cast` on a `linarith`-derived real equation.
- The predicate equivalences are `eq_div_iff/div_eq_iff` followed by `push_cast` and `constructor <;> intro h <;> linarith`.
- The whole unit compiled at the second attempt (the only failure was the name clash).